

4. Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.

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[9] ✓ Os  from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
# Load Iris dataset
iris = load_iris()
X = iris.data
y = iris.target
# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)
# Scale the data
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
# Create ANN model
model = MLPClassifier(
    hidden_layer_sizes=(5,),
    max_iter=2000,
    random_state=1
)
# Train the model
model.fit(X_train, y_train)
# Predict
y_pred = model.predict(X_test)
# Display results
print("Predicted Values:")
print(y_pred)
print("\nActual Values:")
print(y_test)
```

OUTPUT:

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▼ ... Predicted Values:
[0 1 1 0 2 1 2 0 0 2 1 0 2 1 1 0 1 1 0 0 1 1 2 0 2 1 0 0 1 2 1 2 1 2 2 0 1
 0 1 2 2 0 1 2 1]

Actual Values:
[0 1 1 0 2 1 2 0 0 2 1 0 2 1 1 0 1 1 0 0 1 1 1 0 2 1 0 0 1 2 1 2 1 2 2 0 1
 0 1 2 2 0 2 2 1]

Accuracy: 95.55555555555556
```